

Lab 3: Architectural Pattern Selection & Justification

SAIKAVIN S – PES1UG24CS402

Problem Statement — Digital Gift Card & Loyalty Point Exchange System

Architecture Selection: We chose the Layered Architecture for the Digital Gift Card & Loyalty Point Exchange System.

ARCHITECTURAL CHOICE

The system is divided into three layers with clear separation of responsibilities — the Presentation Layer (User Interface Component), the Business Layer (Exchange Manager, Fraud Screening, and Merchant Integration Components), and the Data Layer (Ledger Database Component).

The Exchange Manager is the central business component. It provides `IXchangeRequest` to the User Interface and requires `IMerchantSync` from Merchant Integration, `IFraudCheck` from Fraud Screening, and `ILedgerStore` from the Ledger Database. The providing component shows the ball and the consuming component shows the socket.

TWO REASONS FOR THIS CHOICE

1. Separation of Concerns

The layered structure separates user interaction, business processing, and data management into clearly defined responsibilities. The User Interface handles user interaction, including browsing the catalog (FR-004), displaying aggregated balances, and submitting conversion, redemption, and rate-update requests.

The Business Layer handles aggregation of merchant balances (FR-002), conversion using current merchant-specific rates (FR-003), gift-card redemption (UC-04), voucher generation (UC-05, FR-001), voucher locking (FR-005), fraud screening (UC-06), and merchant integration.

The Data Layer is responsible for persistent ledger, voucher, catalog, rate, and linked-account data. This separation prevents responsibilities from being mixed across the system and makes the architecture easier to understand and manage.

2. Maintainability and Modularity

Because responsibilities are separated and components communicate through defined interfaces, individual components can be modified or replaced without changing the entire system.

For example, the Merchant Integration component can change how it connects to Merchant Partners, and fraud-screening rules for UC-06 can evolve without affecting the User Interface or Ledger Database, provided the defined interfaces remain compatible. Similarly, the database technology can be replaced behind `ILedgerStore` without requiring major changes to the Business or Presentation Layers.

SECURITY ADVANTAGE

Sensitive operations such as redemption, voucher generation, fraud screening, and ledger updates are handled within the Business and Data Layers rather than directly by the Presentation Layer.

The User Interface accesses business functionality through `IXchangeRequest`, while the Exchange Manager controls the business operations. Suspicious redemptions are checked through `IFraudCheck` and can be placed on HOLD before credits are finally deducted, as specified by UC-06 extending UC-04.

This separation limits direct access to sensitive operations and ensures that critical business and data operations are controlled by the appropriate layers.

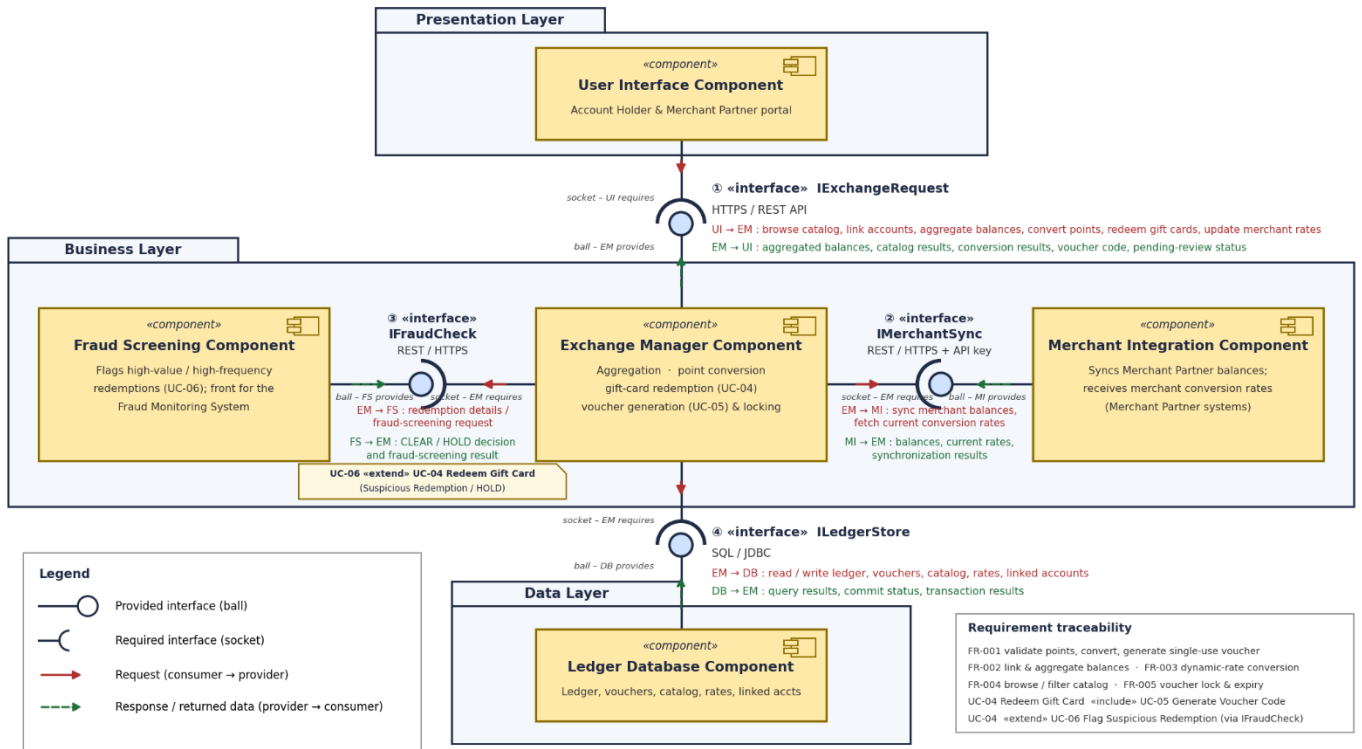
PERFORMANCE BENEFIT

The layered structure keeps business processing centralized in the Exchange Manager and allows direct service-to-service communication between Business Layer components and the Ledger Database.

Merchant balances and current rates can be retrieved through `IMerchantSync`, while ledger operations use `ILedgerStore`, avoiding unnecessary interaction with the Presentation Layer. This supports efficient conversion and redemption processing and helps the system meet the 250 ms performance target specified in NFR-001.

Digital Gift Card & Loyalty Point Exchange System - Layered Architecture

UML Component Diagram | 5 components, 4 interfaces (Problem Statement #34)



PES1UG24CS402 SAIKAVIN S